

proof of Theorem 2.

$$\begin{aligned}\mathcal{B}_1^t = & T_{b,K}^{t-1} + l_K^t \times \left(T_{f,K}^{t-1} + T_{b,K}^{t-1} \right) + T_{f,K}^t + T_a^t + T_{s2c,k}^t \\ & - T_{s2c,k}^{t-1} - T_{b,k}^{t-1} - \bar{l}_k^t \times \left(T_{f,k}^{t-1} + T_{b,k}^{t-1} \right) - T_{f,k}^t,\end{aligned}$$

$$\hat{\mathcal{B}}^t - \mathcal{B}_1^t = \mathcal{G}^t - \left(T_{b,K}^{t-1} + l_K^t \times \left(T_{f,K}^{t-1} + T_{b,K}^{t-1} \right) + T_{f,K}^t \right) > 0.$$

The bubble happens when the active party sends the t -th round signal to the slowest passive party k . The slowest party is still calculating in the $t - B_k^{\tilde{t}}$ round, where $\tilde{t} + B_k^{\tilde{t}} = t$. The embedding used in round t is sent by passive party k in round $\tilde{t}$. The time for the passive party sending the embedding in round t can be defined as:

$$\Lambda^t = \max_i \left(T_{s2c,i}^{\tilde{t}} + T_{f,i}^{\tilde{t}-1} + T_{b,i}^{\tilde{t}-1} + T_{f,i}^t + T_{c2s,i}^t \right).$$

If the max training times $\bar{l}_i^t$ is set to 0, then:

$$\Lambda^t = \max_i \left(T_{s2c,i}^{\tilde{t}} + T_{f,i}^t + T_{c2s,i}^t \right).$$

The time for the active party to receive embeddings from the slowest passive party after sending the gradients for the t -th round is:

$$\begin{aligned}T_{\text{active}}^t = & \Lambda^t - \left(T_{b,K}^{\tilde{t}} + \sum_{j=\tilde{t}+1}^{t-1} \left(T_{f,K}^j + T_a^j + T_{b,K}^j + T_{\text{active}}^j \right) \right. \\ & + \sum_{j=\tilde{t}}^{t-2} \left(l_K^j \times \left(T_{f,K}^j + T_{b,K}^j \right) \right) \\ & \left. + \bar{l}_k^{t-1} \times \left(T_{f,K}^{t-1} + T_{b,K}^{t-1} \right) + T_{f,K}^t \right).\end{aligned}$$

Then we can give the definition of bubble time in case3:

$$\begin{aligned}\mathcal{B}_2^t = & T_{b,K}^{t-1} + \bar{l}_K^{t-1} \times \left(T_{f,K}^{t-1} + T_{b,K}^{t-1} \right) \\ & + T_{f,K}^t + T_{\text{active}}^t + T_a^t + T_{s2c,k}^t \\ & - T_{b,k}^{t-1} - \bar{l}_k^t \times \left(T_{f,k}^{t-1} + T_{b,k}^{t-1} \right) - T_{f,k}^{t+1} - T_{s2c,k}^{t-1}.\end{aligned}$$

The bubble gap between FedBCD and PASTA is:

$$\begin{aligned}T_{B_k^{\tilde{t}}}^t = & \sum_{j=\tilde{t}+1}^t \left(\hat{\mathcal{B}}^j - \mathcal{B}_2^j \right) \\ = & \sum_{j=\tilde{t}+1}^{t-1} \left(T_{f,K}^j + T_a^j + T_{b,K}^j \right) \\ & + \sum_{j=\tilde{t}}^{t-2} \left(l_K^j \times \left(T_{f,K}^j + T_{b,K}^j \right) \right) \\ & + \mathcal{G}^t - \Lambda_{B_i}^t \\ & + \left(B_k^{\tilde{t}} - 1 \right) \times \left(\mathcal{G}^t - V \right) > 0,\end{aligned}$$

where

$$V = \left(T_{b,K}^{t-1} + \bar{l}_K^{t-1} \times \left(T_{f,K}^{t-1} + T_{b,K}^{t-1} \right) \right) + T_{f,K}^t.$$

The total optimization time is the sum of three types of bubbles gap:

$$\Delta \mathcal{B}_k = \sum_{t \in \mathcal{T}_1} \hat{\mathcal{B}}_k^t + \sum_{t \in \mathcal{T}_2} \left(\hat{\mathcal{B}}_k^t - \mathcal{B}_{1,k}^t \right) + \sum_{t \in \mathcal{T}_3} T_{B_k^{\tilde{t}}}^t,$$

where $\mathcal{I}$ is a round indices set of indices that help calculate the bubble. Specifically, if $t \in \mathcal{T}_3$, then $t-1, \dots, t-B_k^{\tilde{t}'} \notin \mathcal{I}$. $\mathcal{T}_1$, $\mathcal{T}_2$ and $\mathcal{T}_3$ are subsets of $\mathcal{I}$, each representing the three

different cases mentioned earlier and ensures that $|\mathcal{T}_1| + |\mathcal{T}_2| + \sum_{t \in \mathcal{T}_3} B_k^{\tilde{t}} = R$. $\square$